FactOWL: A Cost-Efficient Tool for Long-Form Factuality Evaluation

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Abstract

Recent years have seen increasing interest in assessing the factuality of Large Language Models' (LLMs) long-form generations, driven by a severe problem of factual hallucinations. A prominent approach to this challenge is the extract-then-verify framework, which decomposes long-form responses into a set of atomic facts and verifies each using external evidence. Among these methods, FActScore (Min et al., 2023a) stands out as a foundational and widely adopted. However, the research still lacks a unified open-access, up-to-date and cost-effective factuality evaluation tool since FActScore relies on outdated Wikipedia dump and proprietary LLMs available via paid API. To fill this gap, we propose **FactOWL**, a FActScore-based **Fact**uality evaluation tool which adopts an **O**pen LLM and real-time **W**ikipedia search for evaluation of **L**ong-form LLM responses. The proposed tool effectively addresses the problems of obsolete knowledge, incomplete contexts, and entity ambiguity by performing multi-page context aggregation and supporting additional sources, e.g., passages generated from Wikidata triples while showing about a 10x speed-up compared to FActScore. Experiments on FActScore's manually annotated data indicates that FactOWL's scores are close to human-judged factuality scores. Our tool is freely available at: `https://github.com/s-nlp/factowl` under the MIT license and can be installed via *pip install factowl*.

1 Introduction

Even the most advanced LLMs produce factual errors. Unlike LLM hallucinations, when the generation appears meaningless, incoherent, or irrelevant to the prompt, factual errors are much harder to detect (Augenstein et al., 2024; Wang et al., 2024). Factual errors can mislead the user and cause unpleasant consequences. Therefore, methods for assessing the factuality of LLMs attract much attention. The limits of LLM knowledge can be assessed using factoid questions and short golden answers, but this approach is not suitable for evaluating long-form LLM generation. The extract-then-verify approach is usually employed in this case: isolated facts are extracted from the LLM response, external evidence is sought for each of them, and a veracity of the fact is assessed based on the evidence.

FActScore is one of the early implementations of this approach to evaluating LLM's long-form responses (Min et al., 2023a). FActScore uses Wikipedia as a source of evidence; extraction of atomic facts and their assessment is performed using LLMs. Since the publication of this method, it has been actively used in research and practice. However, FActScore has several drawbacks: 1) low performance, 2) the use of three different LMs in the pipeline, one of which (InstructGPT) was available through API and is no longer supported, 3) the use of a static collection of Wikipedia from 2023, 4) an implicit assumption that the entity name can be used as a key to retrieve a single Wikipedia page from the database as evidence. The last assumption is not obvious at first glance, but significantly limits the applicability of the method, especially for less popular entities. For example, the FActScore's explored LLM generation for a lottery winner *William Post*. As seen from Table 2, English Wikipedia provides three people bearing this name – an actor, a businessman, and a lottery winner. While FActScore's static Wikipedia dump returned the ground truth page as top-1 with no disambiguation required, it is not the case for real-world context retrieval where the ground truth is only retrieved as top-3.

In this paper, we present FactOWL, an improved version of FActScore that uses the same conceptual approach and architecture but addresses the issues mentioned above. First, we use a newer open model Llama-3-8B, both to extract atomic facts and assess them against evidence from Wikipedia. This significantly simplifies the pipeline and its

Aspect	FactScore	FactOWL
Fact Extraction	**(closed-weights, deprecated)** InstructGPT	**(open-weights)** LLaMA-3-8B-Instruct
Verification Model	**Old** LLaMA-1-ins	**New** LLaMA-3-8B-Instruct
Knowledge Source	**Static** Wikipedia dump (2023)	**Live** Wikipedia API + optional Wikidata
Entity Disambiguation	**One-to-one** Wikipedia page lookup	**Multi-page** retrieval + reranking
Pipeline Speed	**Slow**, Multi-model	**Fast** Unified model, with vLLM
Availability	**Paid, Proprietary** APIs (InstructGPT)	**Free, Open-Access**
Context Aggregation	**Single-page** top-k paragraphs	**Multi-page**, multi-passage aggregation

Table 1: Comparison of FactScore and FactOWL, highlighting FactOWL's advantages.

maintenance. Second, we use Wikipedia search to find evidence, not just a single Wikipedia page. This, in turn, allows us to address the issue of data freshness and entity name ambiguity. At the same time, unlike more recent alternatives, our approach is cost-efficient: it does not use any commercial search and LLM APIs. FactOWL is freely available either at Github[1] and as a pip package[2].

2 FactOWL

In this section, we describe FactOWL, an open cost-efficient tool for factuality evaluation in long-form generations. FActScore (Min et al., 2023a) was a pioneering work aimed at the factuality evaluation of long-form LLM responses. While pioneering, it has three key limitations we address: (1) static Wikipedia 2023 knowledge, (2) proprietary components, and (3) unrealistic entity disambiguation assumptions. Our methodology relies on FActScore core framework while addressing its key limitations. Following FActScore, *FactOWL* decomposes the factual evaluation into three steps: (i) Splitting an input long-form generation into a set of *atomic facts* followed by (ii) evidence retrieval and (iii) fact verification. FactOWL provides an open, cost-efficient alternative using Llama-3-8B and real-time Wikipedia retrieval.

2.1 FActScore Limitations

Knowledge Reliability As shown in prior work (Wadden et al., 2022; Min et al., 2023a) a factuality estimation is strongly dependent on the knowledge source choice. The original FActScore implementation only supports a static Wikipedia 2023 making the method unreliable and outdated.

Entity Ambiguity While the FActScore's experimental set-up provides an one-to-one exact match between a generation topic and a page from their local Wikipedia 2023 dump, it is not the case in real-world applications. Moreover, FActScore's ground truth supporting page is not always a top-1 result from the current Wikipedia search, as shown in Table 2. Here, a ground truth lottery winner William Post is retrieved as the top-3 result by Wikipedia search and top-1 page retrieval only provides misleading non-relevant context.

Non-Transparency and Costs FActScore adopts proprietary InstructGPT (Ouyang et al., 2022) model to split an input long-form generation into a set of atomic facts, limiting large-scale free and open research on LLM factuality. Follow-up extensions address the FActScore's knowledge source incompleteness and entity ambiguity in single-page set-up by referring to commercial search engines (Zhao et al., 2024; Wei et al., 2024; Song et al., 2024) which further make the inference on new data costly.

2.2 Fact Generation and Verification

LLM Prompting FactOWL prompts an LLM to break down a long-form LLM generation q on a selected topic t into a set of short *atomic facts* $\mathcal{A}_q$. Then, each fact $a \in \mathcal{A}_q$ is independently verified using fact-specific evidence C_a retrieved from an external knowledge base, such as Wikipedia. The verification step is formulated as a binary classification task: given an atomic fact and k retrieved passages, an LLM is required to output whether the fact is supported by the passages. For prompt details, see Appx. D. Overall, factual precision for q is formalized as the ratio of supported facts:

$$f(q) = \frac{1}{|\mathcal{A}_q|} \sum_{a \in \mathcal{A}_q} verify(a, C_a),$$

where $verify(a, C_a)$ is a binary label returned by a verifier LLM: $verify(a, C_a) = 1$ if C_a supports a and $verify(a, C_a) = 0$ otherwise.

[1] https://github.com/s-nlp/factowl
[2] https://pypi.org/project/factowl/

Topic	LLM Generation	Wikipedia search
William Post	William Post (1949-1986) was an American lottery winner who won $16.2 million in the Pennsylvania Lottery in 1988...	1. [C. W. Post] Charles William Post (October 26, 1854 – May 9, 1914) was an American innovator, breakfast cereal and ... 2. [William Post (businessman)] William Post (June 27, 1927 – February 10, 2024) was an American businessman and inventor ... 3. **[William Post (lottery winner)]** William "Bud" Post III (April 5, 1939 – January 15, 2006) was the winner of a Pennsylvania Lottery jackpot worth $16.2 million...

Table 2: Wikipedia search example for *William Post* entity from FActScore dataset. Page titles are provided in square brackets and the ground truth page is highlighted in **bold**.

Model	InstructGPT		ChatGPT		PerplexityAI	
	P	**ER**	**P**	**ER**	**P**	**ER**
Entity-level end-to-end Fact Generation & Verification						
Human evaluation (Min et al., 2023a)	42.5	—	58.3	—	71.5	—
Original FActScore (Min et al., 2023a)	41.1	1.4	58.7	0.4	71.6	0.1
FactOWL, Llama-3-8b-inst+Llama-1, 1 page, 2023 Wikipedia dump	43.7	1.2	53.7	4.6	58.4	13.1
FactOWL (1 page, 2023 Wikipedia dump)	43.6	1.1	59.4	1.1	64.1	7.4
FactOWL (1 page, 5 passages)	40.8	1.7	56.4	1.9	62.2	9.3
FactOWL (5 pages, 10 passages)	43.0	0.5	58.4	0.1	67.9	3.6
FactOWL (5 pages + Wikidata)	44.0	1.5	60.7	2.7	66.8	4.7
Claim-level Atomic Fact Verification						
Human evaluation (Min et al., 2023a)	44.4	—	58.9	—	81.7	—
FactOWL (1 page, 5 passages, 2023 Wikipedia dump)	46.1	1.7	62.5	3.6	72.6	9.1
FactOWL (1 page, 5 passages)	43.7	0.7	57.9	1.0	71.3	10.4
FactOWL (5 pages, 10 passages)	46.8	0.4	61.9	3.0	74.8	6.9
FactOWL (5 pages + Wikidata)	48.1	3.7	62.8	3.9	74.8	6.9

Table 3: FactOWL's factual Precision (**P**) scores on InstructGPT, ChatGPT, and PerplexityAI generations, evaluated against human annotations from FActScore and results produced by the original FActScore pipeline. Results include (i) entity-averaged end-to-end fact generation and verification metrics, and (ii) atomic fact-level metrics based on FActScore's extracted facts. Error Rate (**ER**) denotes the difference compared to manual evaluation.

Model	Total	Disagreements	Disagree, %	TP	TN	FP	FN
Perplexity	5,888	1,108	18.8%	4,052	728	759	349
ChatGPT	5,426	1,047	19.3%	2,773	1,606	433	614
InstructGPT	4,726	700	14.8%	1,819	2,207	285	415

Table 4: Detailed comparison of human and model factscore agreement across Perplexity, ChatGPT and InstructGPT. **True Positives** (**TP**) and **True Negatives** (**TN**) indicate correct alignment with human labels. **False Positives** (**FP**, model supports, humans do not) suggest liberal bias; **False Negatives** (**FN**, model rejects, humans support) suggest conservative bias. InstructGPT shows the lowest disagreement rate and most balanced behavior.

Statistics	InstGPT	ChatGPT	PPLAI
FActScore			
# Total facts	4,726	5,426	5,888
Avg. facts per response	26	34.6	35.46
Inference time	∼2h	∼2h 10m	∼2h 30m
FactOWL			
# Total facts	3,752	4,987	6,401
Avg. facts per response	20.6	31.8	38.6
Inference time	24m 30s	24m 6s	27m 36s

Table 5: Efficiency of FActScore vs. FactOWL (single-page Wikipedia retrieval).

Key Optimizations For fact generation and improvement, we introduce three key improvements over FActScore (Min et al., 2023a). First, we use a single open-source Llama-3-8B-Instruct[3] model (AI@Meta, 2024) for both fact generation and verification replacing a two-model pipeline that relied on proprietary InstructGPT (Ouyang et al., 2022) and Llama-1-7B (Touvron et al., 2023). Second, the superior performance of Llama-3-8B-Instruct eliminates the need for an additional fact

[3] https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct

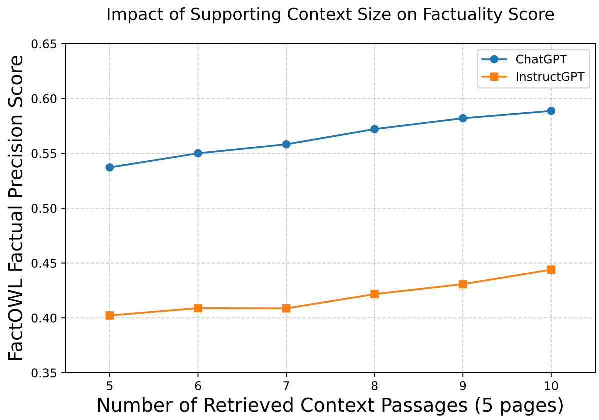

Figure 1: Entity-level comparison of FactOWL to FActScore's manual precision.

filtering step based on masked language modeling (Min et al., 2023b), resulting in faster and simpler inference. Third, we integrate vLLM (Kwon et al., 2023) to further accelerate inference. In general, the adjustments make FactOWL a freely accessible and time-efficient tool for research on LLMs' factuality assessment.

2.3 Supporting Context

Supporting Passages Retrieval While FactOWL is compatible with the original FActScore setup that relies on a static 2023 Wikipedia dump, it also addresses the issue of outdated knowledge by retrieving up-to-date pages through Wikipedia search API. For each topic in a given set of LLM response topics, we retrieve up to the top N pages returned by Wikipedia library.[4] Then, a page is split into a set of sequential textual passages with each passage being no longer than 256 tokens from a single Wikipedia page's section. Furthermore, FactOWL supports the use of arbitrary pre-computed passages from external sources; for instance, additional context from Wikidata[5] can be provided as supporting evidence. Illustrative examples for Wikipedia search results are shown in Table 2.

Thus, FactOWL addresses the problem of entity ambiguity by aggregating content of multiple pages sharing similar or matching names. The outdated knowledge problem is solved by referring to up-to-date pages rather than a static Wikipedia dump. Additionally, FactOWL can use additional passages from any external knowledge source, e.g., Wikidata or a domain-specfic knowledge base.

Passage Reranking As in FActScore, we re-rank all retrieved passages to obtain top k using Generalizable T5-based Retriever (GTR)[6] (Ni et al., 2022). For each passage, we define its title as the concatenation of Wikipedia entity name and section header. In LLM prompt, we concatenate passage title with passage text. An illustrative example is provided in Appx. D.

3 Experiments

In our experiments, we evaluate FactOWL along two main dimensions: (i) its performance on existing benchmarks for factuality evaluation, particularly on the manually annotated LLM generations from FActScore, and (ii) its sensitivity to key configuration parameters, including retrieval source, the number of retrieved pages N, and the number of provided supporting passages k.

3.1 Experimental Setup

Evaluation Data *FActScore* provides Instruct GPT, ChatGPT, and PerplexityAI generations for 183 persons, each aligned with a Wikipedia page as a supporting context. All generations from three LLMs are automatically split into atomic facts, which are subsequently refined and judged by annotators.

Comparison with Human Evaluation To investigate how well FactOWL's factual precision metric aligns with human judgment, we use manually labeled data from FActScore in two evaluation setups: (i) **Fact Extraction & Verification**, where we compute end-to-end precision scores averaged

[4] https://pypi.org/project/wikipedia/
[5] https://www.wikidata.org

[6] https://huggingface.co/sentence-transformers/gtr-t5-large

over entities, and (ii) **Fact Verification**, based manually revised atomic facts from FActScore. Task (i) evaluates how well the overall precision score of a long-form generation reflects human assessments, while task (ii) explores the difference between manual and automatic fact verification exclusively.

Implementation Details In our experiments, we also evaluate two-model FactOWL set-up with Llama-3-8b for fact generation and Llama-1-inst for fact verification. For single- and multi-page setting, we experiment with top 5 and top 10 retrieved passages for each atomic fact, respectively.

Wikidata Passages FactChecker can be enhanced with additional knowledge by incorporating passages derived from Wikidata through the following steps: 1) **Identify the Wikidata ID** of the target entity via its Wikipedia URL using a SPARQL query; 2) **Retrieve all triples** where the entity is the subject or object using SPARQL; 3) **Convert triples to natural language** using templates generated per relation type via the Meta-Llama-3.1-70B-Instruct model, inserting the subject and object into each template; 4) **Aggregate sentences into passages** by relation type, using the relation as the passage topic. Template prompts and examples are provided Appendix E.

3.2 Results

Table 3 compares the FactOWL's automatic factual precision to manual annotation on FActScore data.

FactOWL is Close to Human Evaluation As seen from the results, our tool shows factuality precision scores close to human evaluation for InstructGPT and ChatGPT for both end-to-end task and fact verification only set-ups. While FactOWL shows lower gap to Human evaluation that the original FActScore implementation on end-to-end evaluation from 1.4 to 0.5 for InstructGPT and from 0.4 to 0.1 for ChatGPT, the gap increases for Perplexity generations from 0.1 to 3.6. We will further investigate the issue in Sec. 4.

More Evidence Improves Precision FactOWL consistently benefits from more supporting documents available. For *fact verification*, the precision score increases by 3.1, 4.0, and 2.5 on InstructGPT, ChatGPT, and PerplexityAI, respectively, when five supporting pages are retrieved instead of a single page. It further increases by 1.3 and 0.9 for InstructGPT and ChatGPT, respectively, when additional Wikidata passages are provided. Overall, the similar results are observed for end-to-end evaluation with the only exception of Perplexity showing the highest precision without Wikidata context.

Inference Time Table 5 provides results on prediction time and retrieved facts count. As seen from the results, FactOWL extracts comparable number of facts while showing about $6x$ inference time speed-up.

Larger Context Increases Precision Figure 1 shows the FactOWL's factuality metric change over different number of supporting passages. For 5-page retrieval results, larger context consistently increases the precision score. The finding indicates that the best-matching supporting passage is not always retrieved among top-5 passages which is caused by retrieved entities having similar names (entity ambiguity) and thus being harder to distinguish.

4 Analysis

Human vs LLM Evaluation Alignment A detailed analysis of the alignment between three large language models (LLMs) and human evaluations is presented in Table 4. All three models' generations align with human labels in the majority of cases, with InstructGPT demonstrating the highest agreement rate (85.2%), suggesting stronger alignment with factual judgments. Both ChatGPT and InstructGPT tend to produce more false negatives than false positives, indicating a more conservative approach to factuality. In contrast, Perplexity exhibits a more liberal pattern, generating a higher number of false positives.

The True Positives count is highest for Perplexity, which may reflect an optimistic bias toward accepting claims as factual. Meanwhile, True Negatives (correct rejections) are most frequent with InstructGPT, suggesting a more cautious and critical evaluation style. Overall, InstructGPT's lower disagreement rate and conservative behavior indicate that it may be better suited for fact-checking tasks in high-stakes or safety-critical domains.

Case study. Table 6 presents a case study focused on the entity *Shahnaz Pahlavi*, where we selected all atomic claims for which human evaluation and the FactOWL disagreed on Perplexity generations. We then manually re-evaluated these claims by consulting three relevant Wikipedia articles. For each

Atomic claim	Human	FactOWL	Wiki	Proof from Wikipedia
King Faisal was the King of Iraq.	✗	✓	✓	"Faisal II ... was the last King of Iraq." *Introductory sentence* in [1]
Their son was called Keykhosrow.	✗	✓	✓	"They have a son, Keykhosrow (born 20 November 1971)" *Personal life section* in [2]
Their daughter was called Fawzia.	✗	✓	✓	"...and a daughter, Fawzia (born 1973)" *Personal life section* in [2]
Fawzia was born in 1973.	✗	✓	✓	"...and a daughter, Fawzia (born 1973)" *Personal life section* in [2]
Shahnaz's mother-in-law was Queen Fawzia.	✓	✗	✗	"She is the only child of Mohammad Reza Pahlavi and Queen Fawzia." *Personal life section* in [2]
Queen Fawzia was a Queen of Egypt.	✓	✗	✗	"Fawzia of Egypt was an Egyptian princess who became Queen of Iran." *Introductory section* in [3]
Mohammad Reza Shah Pahlavi and she had one child.	✓	✗	✓	"...she had one child, a daughter." *Queen of Iran section* in [3]
Their child was a daughter.	✓	✗	✓	"...she had one child, a daughter." *Queen of Iran section* in [3]
Their daughter is HIH Princess Shahnaz Pahlavi.	✓	✗	✓	"HIH Princess Shahnaz Pahlavi (born 27 October 1940)." *Queen of Iran section* in [3]
HIH Princess Shahnaz Pahlavi was born on October 27, 1940.	✓	✗	✓	"HIH Princess Shahnaz Pahlavi (born 27 October 1940)." *Queen of Iran section* in [3]

[1] https://en.wikipedia.org/wiki/Faisal_II
[2] https://en.wikipedia.org/wiki/Shahnaz_Pahlavi
[3] https://en.wikipedia.org/wiki/Fawzia_of_Egypt

Table 6: Claims about *Shahnaz Pahlavi* that did not match between the human evaluation reported in the original FactScore and FactOWL were double-checked against the corresponding Wikipedia articles. The "Proof from Wikipedia" column indicates the specific sections within these articles where each claim can be verified. Numerical references point to source articles. In 6/10 cases human labelling was wrong while FactOWL was correct.

claim, we provide a reference to the specific section in the source where it can be verified.

Our analysis revealed that the initial human labeling was correct in only 4 out of 10 cases, while the FactOWL checker provided the correct judgment in 6 out of 10. While this is a small sample, it offers a cautiously optimistic indication that automated fact-checking using language models may outperform human judgments in certain contexts.

5 Conclusion

We introduce FactOWL, a novel lightweight tool for assessing factual precision of long-form LLM generations. Compared to prior research, it investigates the entity ambiguity and outdated knowledge problems by using multi-page real-time Wikipedia search and supporting external pre-computed knowledge passages (e.g., verbalized Wikidata triples). Experiments have shown that our method's estimated factuality precision is consistent with human judgements on LLM generations from FActScore. By relying on non-proprietary Llama-3-8B-Instruct model and an open-access Wikipedia search tools, FactOWL encourages open research on LLM factuality.

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A Related Work

Search-Augmented Factuality Evaluator (SAFE) (Wei et al., 2024) can be seen as an extension of FActScore in the part of evidence search. After extracting atomic facts in the same way as FActScore, SAFE iteratively searches the Web for evidence: for each atomic fact, the method makes five search API calls and five LLM calls to generate search queries. The authors estimate that evaluating a single LLM response costs $0.2–0.4, taking into account the cost of the Serper and OpenAI API calls. This is significantly less than the cost of manual evaluation, but still may be prohibitive for small teams to conduct evaluation studies involving several LLMs in different configurations using modern standard datasets containing thousands of test instances (e.g., LongFact (Wei et al., 2024) contains 2,280 test prompts).

In contrast to SAFE, DnDScore (Wanner et al., 2024) and Claimify (Metropolitansky and Larson, 2025) focus on the initial stage of fact-checking: claim extraction. Specifically, these studies discuss the desired properties of claims, such as atomicity, entailment, coverage, and self-containedness and propose LLM-based pipelines to fulfill these criteria.

VeriScore's (Song et al., 2024) approach is similar to FActtScore and SAFE, but it specifically aims to extract *verifiable claims*. It uses Google search results as the source of evidence. In line with our efforts to provide a cost-efficient fact-checking solution, VeriScore has a version that uses an open LLM fine-tuned on data generated by closed GPT models.

B Hyper-parameters

For fact generation, vLLM parameters for meta-llama/Meta-Llama-3-8B-Instruct were as follows: $max\_new\_tokens = 2048$, $temperature = 0.0$. For fact verification, we set $max\_new\_tokens = 8$ as the model needs to predict either True or False.

C Hardware set-up

All experiments were conducted on a server equipped with 2× AMD EPYC 7763 64-Core CPUs (256 threads total), 3 NVIDIA RTX-3090 GPU (24 GB memory each), and sufficient system RAM to support large-scale computation.

D LLM prompts

LLM system prompt for facts extraction.

*You are an **expert fact extraction** and **verification assistant**.*

Please read the following text carefully and break it down into distinct, independent facts. For each fact, disambiguate it to ensure clarity and precision (e.g., replace ambiguous prepositions).

Each fact should be written on its own line. Each line must start with a hyphen and space ('- '). Do not include any additional explanation or formatting - just the list of facts if there are any.

LLM system prompt for facts verification.

*You are an **expert fact extraction** and **verification assistant**.*

*You are given an atomic fact and a list of textual passages. Your task is to determine whether the atomic fact is **True** or **False** based solely on the information in the passages.*

Instructions:

1. *Check if any of the passages directly support the atomic fact.*
2. *Output **'True'** if at least one passage supports the fact even if another passage contradicts the fact.*
3. *Output **'False'** if no passage supports the fact.*
4. *Do not include any additional information and explanations, you must only answer **'True'** or **'False'**.*

Fact verification query template.

Passages:
<passages>
Atomic Fact's topic/topics: *<fact_topic>*
Atomic Fact: *<atomic_fact>*
True or False?

E Wikidata passages collection

System prompt:

You are a knowledgeable language assistant that transforms Wikidata triples (subject predicate, object) into fluent, contextually appropriate English sentences. You must vary the sentence structure naturally based on the meaning of the relation and ensure grammatical accuracy and idiomatic phrasing.

User prompt:

Technical details. Generation parameters for meta-llama/Meta-Llama-3.1-70B-Instruct were as follows: $max\_new\_tokens = 128$, $temperature = 0.3$, $top\_p = 0.9$.

Convert this Wikidata triple into a clear and natural English sentence. The sentence must be grammatically correct and reflect the meaning of the relation accurately.

Examples:
['Lanny Flaherty', 'place of death', 'New York City'] → Lanny Flaherty died in New York City.
['Marianne McAndrew', 'native language', 'English'] → Marianne McAndrew's native language is English.
['Melrose Place', 'cast member', 'Andrew Shue'] → Andrew Shue is a cast member of the soap opera "Melrose Place".
['2nd G7 summit', 'participant', 'Takeo Miki'] → Takeo Miki was a participant in the 2nd G7 summit.

Use the natural subject-object order based on the meaning of the relation. For example, the person is the subject in "native language" or "place of death," while for "participant" or "cast member," the person is the object.

Now convert the following triple:

Examples. Some examples created for *Lanny Flaherty.*

- *topic:* 'cast member', *text*: "Lanny Flaherty is a cast member of Signs. Lanny Flaherty is a cast member of Miller's Crossing. Lanny Flaherty is a cast member of Waterworld..."
- *topic:* 'place of birth', *text*: "Lanny Flaherty was born in Pontotoc."
- *topic:* 'country of citizenship', *text*: "Lanny Flaherty is a citizen of United States."